

Yurina Nakazato

Curriculum Vitae (last updated December 2025)

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Research Interests

Galaxy formation and evolution, cosmology, first stars and galaxies, cosmological simulations, multi-wavelength observations

Work Experience

Oct. 2025 - present **Flatiron Research Fellow, Center for Computational Astrophysics**, Flatiron Institute, New York, US

Education

Apr. 2023 - Sep. 2025 Ph.D. student, The University of Tokyo, Japan (Completed ahead of schedule)
Thesis: *Formation and Evolution of High-Redshift Galaxies Using Cosmological Simulations and Multi-Wavelength Observations*
Advisor: Naoki Yoshida

Apr. 2021 - Mar. 2023 M.S. in Physics, The University of Tokyo, Japan
Thesis: *Formation and evolution of star clusters and galaxies in the early Universe*
Advisor: Naoki Yoshida
President's Award (1/3000), Dean's Award (1/128)

Apr. 2017 - Mar. 2021 B.S. in Physics, The University of Tokyo, Japan

Fellowships and Awards

Oct. 2025 The L'Oréal-UNESCO For Women in Science Japan Fellowship

Aug. 2025 The Excellence Award for Final Exam

Jan. 2025 Flatiron Research Fellow

Jan. 2025 CITA Fellow (declined)

Feb. 2024 Outstanding Poster Award at the International Symposium on Quantum Electronics 2024

Apr. 2023- Sep.2025 JSR fellowship

Apr. 2023- Sep.2025 Japan Society for Promotion of Science (JSPS) Research Fellow, DC1

Mar. 2023 The School of Science Encouragement Award

Mar. 2023 The University of Tokyo President's Award

Mar. 2023 The Excellence Award for Qualifying Exam

Oct. 2021- Sep.2025 The International Graduate Program for Excellence in Earth-Space Science (IGPEES), the University of Tokyo

Aug. 2021 Best Oral Presentation Award at 51th astronomical meeting for junior researchers

Leadership & Outreach

Jun. 2023 Organizer of Get-Together Event for Women in STEM, the University of Tokyo

Jul. 2022 Public Talk at Women in STEM, School of Science, the University of Tokyo

Dec. 2019 Rikejo Initiative, the University of Tokyo

Oct. 2019 Invited talk at Dow Chemical Company, Tokyo, Japan

Grants

Apr. 2023-Mar.2026	Evolution of first galaxies with multi-wavelength observations and numerical simulations, 4.2M JPY (30K USD), JSPS Grant-in-Aid for Early-Career Scientists, No. 23KJ0728
Apr. 2023-Mar.2026	Grand Aid from JSR fellowship, 3M JPY (21K USD)

Coverage in Media

Hunting Faint Galaxies with JWST as a Test for the Dark Matter Model based on Williams et al. 2024
- *Bright galaxies put dark matter to the test*, UCLA

The core of the most distant galaxy protocluster observed by JWST & ALMA based on Hashimoto et al. 2023
- *The Strong Tag Team of the James Webb Space Telescope and ALMA Captures the Core of the Most Distant Galaxy Protocluster*, Kavli IPMU, Tsukuba Univ., Waseda Univ., Nagoya Univ.

Teaching & Professional Service

Teaching

Apr. 2021-Aug. 2021	Teaching Assistant of Fluid Mechanics, the University of Tokyo
Apr. 2019- Jan. 2024	Language Assistant, International Lounge, the University of Tokyo

Professional Service

Nov. 2025-present	Referee of Astronomy & Astrophysics (A&A), The Astrophysical Journal (ApJ), The Open Journal of Astrophysics (OJA)
Apr. 2025	Organizer , One-day workshop on for stellar-cluster simulation, The University of Tokyo, Tokyo, Japan
Dec. 2024	Organizer , One-day workshop on high-z cosmological simulations, The University of Tokyo, Tokyo, Japan
Aug. 2023	Organizer , The 53rd Summer School on Astronomy and Astrophysics, The University of Tokyo, Tokyo, Japan
Mar. 2023	Workshop Organizer , Astrophysics Workshop for Young Researchers, The University of Tokyo, Tokyo, Japan

Proposals

co-Investigator on ALMA proposals

4. 2023.1.00843.S, PI: Y.Tamura, 15.6h
Disentangling the ionization structure of HII regions in a density-bounded Ly α emitter in the Reionization Era
3. 2023.1.00193.S, PI: Y. Fudamoto, 17.3h
[CII] 158 μ m emission line and dust observation of the most distant known overdensity of galaxies at $z = 7.88$
2. 2023.1.00235.S, PI: Y. Fudamoto, 10.0 h
The ALMA-JWST synergy: [OIII]88 μ m and [CII]158 μ m emission line observations of $z = 9.51$ galaxy identified by JWST
1. 2023.1.01362.S, PI: Y. Tamura, 25.9 h
Pandora's ELPIS: The Emission-Line Protocluster Imaging Survey of the furthest overdensity beyond Pandora's Cluster

Active collaborations

- Supersonic Project

- FirstLight simulations
- Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA, JWST GO1840)

International Conferences (Recent Highlights)

total: 28 (27 oral talks, 9 peer-reviewed talks)

- Mar. 2026 (keynote) **Star Forming Galaxies in the Reionization Epoch: ALMA-Driven Studies with the Aid of JWST**, Ishigaki, Japan
- Feb. 2026 (invited) **From Dust till Dawn**, Leiden, Netherlands
- Sep. 2024 **ILR workshop 2024 @Osaka: From Galaxy Formation to Supermassive Black Holes**, Osaka, Japan
- Aug. 2024 **IAUS391**, Cape Town, South Africa
- Jul. 2024 **European Astronomical Society (EAS) Annual meeting 2024**, Padova, Italy
- Nov. 2023 **Resolving the Extragalactic Universe with ALMA & JWST**, Waseda Univ., Japan
- Jul. 2023 **Shedding new light on the first billion years of the Universe**, Marseille, France
- Sep. 2022 **The 9th East Asian Numerical Astrophysics Meeting (EANAM9)**, Okinawa, Japan
- Aug. 2022 **Star Formation in Different Environments (SFDE) 2022**, Quy Nhon, Vietnam
- Jul. 2022 **National Astronomy Meeting (NAM) 2022**, online, hosted by the University of Warwick
- Nov. 2021 **IAU Symposium 362: The predictive power of computational astrophysics as a discovery tool**, online

Seminars (Selected)

total: 16 talks

- Apr. 2026 **University of Maryland**, invited seminar, The United States
- Nov. 2025 **Flatiron Institute**, CCA-NY area galaxy seminar, The United States
- Aug. 2025 **Hokkaido University**, invited seminar, Japan
- Jul. 2025 **Kyoto University**, invited seminar, Japan
- Feb. 2025 **Tsukuba University**, invited seminar, Japan
- Apr. 2024 **Scuola Normale Superiore**, seminar, Italy
- Dec. 2024 **Universidad Autónoma de Madrid**, special seminar, Spain
- Dec. 2024 **Centro de Astrobiología, (CAB, CSIC-INTA)**, special seminar, Spain
- Feb. 2023 **UCLA**, invited seminar, The United States

Working experience as a visiting researcher (student)

- Apr. 2024 - Jul. 2024 **Visiting Student at Scuola Normale Superiore di Pisa**
 - Host researcher is Prof. Andrea Ferrara.
 - Worked on radiation-driven dusty outflow at $z > 10$ galaxies.
- Nov. 2023 - Dec. 2023 **Visiting Student at Universidad Autónoma de Madrid**
 - Host researcher is Prof. Daniel Ceverino.
 - Worked on formation and evolution of clumpy galaxies by using FirstLight simulation
- Feb. 2023 - Mar. 2023 **Visiting Student at UCLA**
 - Host researcher is Prof. Smadar Naoz.
 - Worked on Supersonic Project [link]

Competencies

- Scientific programming : Python, C++, Fortran, Latex
- Simulation software : SKIRT, AREPO, ART, RAMSES-RT
- Astrophysics tool : CLOUDY(photoionizaion code), Astrodendro, SKIRT(dust radiative transfer)

References

Prof. Naoki Yoshida (Ph.D. supervisor), The University of Tokyo, naoki.yoshida@ipmu.jp

Prof. Andrea Ferrara, Scuola Normale Superiore, andrea.ferrara@sns.it

Prof. Smadar Naoz, University of California, Los Angeles, snaoz@astro.ucla.edu

Prof. Daniel Ceverino, Universidad Autonoma de Madrid, Madrid, daniel.ceverino@uam.es